Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India
(Autonomous College Affiliated to University of Mumbai)

Mid Semester Examination
September 2026

Max. Marks: 30 Duration: one Hr

Class: TE Semester: VI

Course Code: CECS302 Program: CE/CSE

Name of the Course:- Software Computing

Instruction:
(1) All questions are compulsory and write answer in exact order of questions.
(2) Draw neat diagrams and write equations wherever necessary
(3) Assume suitable data if necessary

Course Outcomes:
(1) Illustrates software development lifecycle models.
(2) Analyze user requirements and specifications to identify functional and non-functional requirements of a software system.
(3) Design structured and object-oriented system using appropriate design patterns.

Q No.		Max. Marks	CO BL
Q.1	**a)** Why is an association class used for a class diagram? Give example of an association class that may be found in a class diagram that captures students and courses they have to take. **4M** **b)** Create a use case diagram to illustrate use cases for a dentist office system. Whenever new patients are seen for the first time, they complete a patient information form that asks their name, address, phone number and brief medical history which is stored in the patient information file. When a patient calls to schedule a new appointment or change an existing appointment, the receptionist checks the appointment file for an available time. Once a good time is found for the patient, the appointment is scheduled. If the patient is a new patient, an incomplete entry is made in the patient file; the full information will be collected when the patient arrives for the appointment. Because appointments are often made far in advance, the receptionist usually mails a reminder postcard to each patient two weeks before his/her appointment. After completing the current appointment, bill is generated and payment accepted. Identify actors, all use cases	10	3-4

	including **include** and **extend** use case. **6M**		
Q.2	a) A government organization is developing a payroll management system. The requirements are clearly defined by the government and are unlikely to change during development. **(8M)** i) Why the Waterfall Model is suitable for this project. Mention any two limitations of using this model. ii) Apply the Waterfall Model to the given scenario and demonstrate how the development team would carry out the phases b) A library automation system allows students to search for books, issue and return books, and pay fines. The librarian maintains book and member records. Draw a suitable Level-1 DFD for the Library Management System and represent the major data stores and data flows.**(2M)**	10	1-3
Q.3	**a)** A company wants to develop an **Online E-Commerce System** where customers can register, browse products, search for products, and add products to a shopping cart. Customers can place orders and make payments online. The system should show product details and product availability. After an order is placed, the customer should receive the product at the given delivery address. Customers should also be able to cancel orders and provide feedback about products. An administrator can add new products, update product information, remove products, and view customer orders. The system should be **user-friendly, secure, fast and reliable**. 1) Identify **4 requirements** in the above description that are ambiguous, incomplete, or unclear. **(2 M)** 2) Identify **2 functional requirements** and **2 non-functional requirements** from the description. **(2 M)** 3) For any 1 functional requirement, write a detailed and formatted use case description **(2 M)**	10	2-4